ASSIGNMENT-1

Course: Full Stack Web Development
Course Code: 23CM4121
Assignment No.: 1
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1. TITLE

Types of Inheritance in JavaScript Using ES6 Classes

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2. AIM

To understand and implement the five types of inheritance in JavaScript — Single, Multilevel, Hierarchical, Multiple (via Mixins), and Hybrid — using ES6 class syntax, the extends keyword, super() constructor calls, and method overriding with super.display().

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3. OBJECTIVE

The objectives of this experiment are:
- To understand the concept of inheritance and how it promotes code reuse in object-oriented programming.
- To use ES6 class syntax to define base classes and derived classes using the extends keyword.
- To call the parent class constructor from a derived class using super() to initialise inherited properties.
- To override methods in derived classes and chain parent behaviour using super.display().
- To implement Single Inheritance where one derived class extends one base class.
- To implement Multilevel Inheritance where a chain of three classes — Vehicle, Car, SportsCar — is formed.
- To implement Hierarchical Inheritance where two derived classes (Car and Bike) both extend one base class (Vehicle).
- To simulate Multiple Inheritance in JavaScript using a Mixin function that combines Vehicle and MusicSystem behaviour into Car.
- To implement Hybrid Inheritance by combining Multilevel and Hierarchical patterns in a single program.

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4. THEORY

Inheritance — a core principle of object-oriented programming that allows a class (derived/child) to acquire properties and methods of another class (base/parent), enabling code reuse and a logical hierarchy.

ES6 Class Syntax — JavaScript ES6 introduced the class keyword as syntactic sugar over prototype-based inheritance. A class body contains a constructor() method for initialisation and other methods that are added to the class prototype.

extends Keyword — used in a class declaration to set up the prototype chain between a derived class and its base class, giving the derived class access to all non-private members of the base class.

super() — called inside a derived class constructor to invoke the parent class constructor and initialise inherited properties. It must be called before accessing this in the derived constructor.

Method Overriding — a derived class defines a method with the same name as one in the base class. The derived version replaces the base version for objects of that class. super.methodName() can still call the original parent method.

Single Inheritance — one derived class (Car) inherits from one base class (Vehicle). The simplest form: Car adds its own property (color) and calls super.display() to chain output.

Multilevel Inheritance — a chain of three classes: Vehicle is the root, Car extends Vehicle, and SportsCar extends Car. Each level adds its own property and chains super.display() through all levels.

Hierarchical Inheritance — two or more derived classes (Car, Bike) independently extend the same base class (Vehicle). Each child class adds its own unique property while sharing Vehicle's constructor and display().

Multiple Inheritance (Mixin) — JavaScript does not support multiple extends. A Mixin is a higher-order function that accepts a base class and returns a new class that extends it with additional methods. class Car extends MusicMixin(Vehicle) combines both.

Hybrid Inheritance — a combination of two or more inheritance types in one program. Here, SportsCar extends Car (Multilevel) while Bike independently extends Vehicle (Hierarchical), forming both patterns simultaneously.

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5. ALGORITHM / PROCEDURE

- Create five separate JavaScript files: single_Inheritance.js, multilevel_Inheritance.js, hierarchical_Inheritance.js, multiple_Inheritance.js, and hybrid_Inheritance.js.
- In each file, define the base class Vehicle with a constructor(make, model, year) and a display() method that logs all three properties.
- Single: define class Car extends Vehicle, add color in the constructor via super(), override display() calling super.display(), create a Car object and call display().
- Multilevel: define Car extends Vehicle with color; define SportsCar extends Car with topSpeed; each display() calls super.display(); create a SportsCar object and call display().

- Hierarchical: define Car extends Vehicle with color and Bike extends Vehicle with type; both override display() with super.display(); create one Car and one Bike object and call display() on each.
- Multiple: define the MusicMixin higher-order function that returns an anonymous class extending BaseClass with a playMusic() method; define Car extends MusicMixin(Vehicle) with color; create a Car object and call both display() and playMusic().
- Hybrid: combine Multilevel and Hierarchical — define Car extends Vehicle, SportsCar extends Car, and Bike extends Vehicle; create a SportsCar and a Bike object and call display() on each.
- Run each file in Node.js or a browser console and verify the console output against expected results.
- Record expected vs. actual output for each test case.

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6. PROGRAM

Program 1: single_Inheritance.js

```
// Single Inheritance

// Base class: Vehicle
class Vehicle {
  constructor(make, model, year) {
    this.make = make;
    this.model = model;
    this.year = year;
  }
  // Method to display vehicle information
  display() {
    console.log(`This is a ${this.year} ${this.make} ${this.model}.`);
  }
}

// Derived class: Car extends Vehicle
class Car extends Vehicle {
  constructor(make, model, year, color) {
    super(make, model, year); // Call the constructor of the base class
    this.color = color;
  }
  // Method to display car information
  display() {
```

```
    super.display(); // Call the display method from the base class
    console.log(`The car is in ${this.color} color.`);
  }
}

const myCar = new Car("Toyota", "Corolla", 2021, "Blue");
myCar.display();
```

Program 2: multilevel_Inheritance.js

```
// Multilevel Inheritance

// Base class: Vehicle
class Vehicle {
  constructor(make, model, year) {
    this.make = make; this.model = model; this.year = year;
  }
  display() {
    console.log(`This is a ${this.year} ${this.make} ${this.model}.`);
  }
}

// Derived class: Car extends Vehicle
class Car extends Vehicle {
  constructor(make, model, year, color) {
    super(make, model, year);
    this.color = color;
  }
  display() {
    super.display();
    console.log(`The car is in ${this.color} color.`);
  }
}

// Derived class: SportsCar extends Car
class SportsCar extends Car {
```

```
  constructor(make, model, year, color, topSpeed) {
    super(make, model, year, color);
    this.topSpeed = topSpeed;
  }
  display() {
    super.display();
    console.log(`The top speed of the sports car is ${this.topSpeed} km/h.`);
  }
}

const mySportsCar = new SportsCar("Ferrari", "488 GTB", 2022, "Red", 330);
mySportsCar.display();
```

Program 3: hierarchical_Inheritance.js

```
// Hierarchical Inheritance

// Base class: Vehicle
class Vehicle {
  constructor(make, model, year) {
    this.make = make; this.model = model; this.year = year;
  }
  display() {
    console.log(`This is a ${this.year} ${this.make} ${this.model}.`);
  }
}

// Derived class: Car extends Vehicle
class Car extends Vehicle {
  constructor(make, model, year, color) {
    super(make, model, year);
    this.color = color;
  }
  display() {
    super.display();
    console.log(`The car is in ${this.color} color.\n`);
```

```
  }
}

// Derived class: Bike extends Vehicle
class Bike extends Vehicle {
  constructor(make, model, year, type) {
    super(make, model, year);
    this.type = type;
  }
  display() {
    super.display();
    console.log(`The bike is a ${this.type} type.\n`);
  }
}

const myCar = new Car("Toyota", "Camry", 2023, "Black");
myCar.display();

const myBike = new Bike("Yamaha", "R15", 2024, "Sports");
myBike.display();
```

Program 4: multiple_Inheritance.js

```
// Multiple Inheritance

// Base class: Vehicle
class Vehicle {
  constructor(make, model, year) {
    this.make = make; this.model = model; this.year = year;
  }
  display() {
    console.log(`This is a ${this.year} ${this.make} ${this.model}.`);
  }
}

// Mixin: MusicSystem functionality
```

```
const MusicMixin = (BaseClass) => {
  return class extends BaseClass {
    playMusic() {
      console.log("The car is playing music.");
    }
  };
};

// Derived class: Car extends Vehicle + MusicMixin
class Car extends MusicMixin(Vehicle) {
  constructor(make, model, year, color) {
    super(make, model, year);
    this.color = color;
  }
  display() {
    super.display();
    console.log(`The car is in ${this.color} color.`);
  }
}

const myCar = new Car("Toyota", "Corolla", 2021, "Blue");
myCar.display();
myCar.playMusic();
```

Program 5: hybrid_Inheritance.js

```
// Hybrid Inheritance

// Base class: Vehicle
class Vehicle {
  constructor(make, model, year) {
    this.make = make; this.model = model; this.year = year;
  }
  display() {
    console.log(`This is a ${this.year} ${this.make} ${this.model}.`);
  }
```

```
}

// Derived class: Car extends Vehicle
class Car extends Vehicle {
  constructor(make, model, year, color) {
    super(make, model, year);
    this.color = color;
  }
  display() {
    super.display();
    console.log(`The car is in ${this.color} color.`);
  }
}

// Derived class: SportsCar extends Car
class SportsCar extends Car {
  constructor(make, model, year, color, topSpeed) {
    super(make, model, year, color);
    this.topSpeed = topSpeed;
  }
  display() {
    super.display();
    console.log(`The top speed is ${this.topSpeed} km/h.\n`);
  }
}

// Derived class: Bike extends Vehicle
class Bike extends Vehicle {
  constructor(make, model, year, type) {
    super(make, model, year);
    this.type = type;
  }
  display() {
    super.display();
    console.log(`The bike is a ${this.type} type.`);
```

```
  }
}

const mySportsCar = new SportsCar("Ferrari", "488 GTB", 2022, "Red", 330);
mySportsCar.display();

const myBike = new Bike("Yamaha", "R15", 2024, "Sports");
myBike.display();
```

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7. CODE EXPLANATION

Code	Explanation
class Vehicle { ... }	Defines the base class Vehicle with a constructor that sets make, model, and year properties and a display() method that logs vehicle information.
class Car extends Vehicle	Declares Car as a derived class that inherits all properties and methods from Vehicle using the extends keyword.
super(make, model, year)	Calls the parent class constructor from inside the derived class constructor, initialising the inherited properties before adding new ones.
super.display()	Calls the display() method of the parent class from inside the overriding display() method in the derived class, so both messages are printed.
this.color / this.type / this.topSpeed	Properties added exclusively by the derived class constructor after calling super(), extending the parent class data.
class SportsCar extends Car	Multilevel and Hybrid: declares a second level of inheritance where SportsCar inherits from Car, which already inherits from Vehicle, forming a chain.
class Bike extends Vehicle	Hierarchical and Hybrid: a second derived class that also extends Vehicle independently of Car, so one base class has multiple child classes.
const MusicMixin = (BaseClass) => { return class extends BaseClass { ... }; }	Multiple Inheritance: a higher-order function (mixin) that takes a class as an argument and returns a new anonymous class that extends it, adding the playMusic() method.
class Car extends MusicMixin(Vehicle)	Multiple Inheritance: applies MusicMixin to Vehicle first, then Car extends the resulting combined class, simulating multiple inheritance in JavaScript.
myCar.playMusic()	Multiple Inheritance: calls the method contributed by the mixin, demonstrating that the Car object has access to both Vehicle methods and MusicMixin methods.

Code	Explanation
new Car(...) / new Bike(...) / new SportsCar(...)	Creates instances of derived classes. Each object carries its own data but inherits shared methods from the prototype chain built by extends.
.display() call on instance	Triggers the overridden display() in the derived class, which internally calls super.display() to chain the output from every level of the inheritance hierarchy.

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8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Run single_Inheritance.js: new Car("Toyota","Corolla",2021,"Blue")	Prints "This is a 2021 Toyota Corolla." then "The car is in Blue color."	Both lines printed correctly	Pass
TC-02	Run multilevel_Inheritance.js: new SportsCar("Ferrari","488 GTB",2022,"Red",330)	Prints Vehicle line, Car colour line, then top speed line in sequence	All three lines printed in correct order	Pass
TC-03	Run hierarchical_Inheritance.js: myCar.display()	Prints "This is a 2023 Toyota Camry." then "The car is in Black color."	Car output printed correctly	Pass
TC-04	Run hierarchical_Inheritance.js: myBike.display()	Prints "This is a 2024 Yamaha R15." then "The bike is a Sports type."	Bike output printed correctly	Pass
TC-05	Run multiple_Inheritance.js: myCar.display()	Prints "This is a 2021 Toyota Corolla." then "The car is in Blue color."	Inherited Vehicle and Car output printed correctly	Pass
TC-06	Run multiple_Inheritance.js: myCar.playMusic()	Prints "The car is playing music."	Mixin method accessed and executed on Car instance	Pass
TC-07	Run hybrid_Inheritance.js: mySportsCar.display()	Prints Vehicle line, Car colour line, then top speed line	Three-level chain printed in correct order	Pass
TC-08	Run hybrid_Inheritance.js: myBike.display()	Prints "This is a 2024 Yamaha R15." then "The bike is a Sports type."	Bike branch printed independently of SportsCar branch	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-09	Verify super() in all files	Parent constructor initialises make, model, year before child adds its own property	Properties set correctly in every instance across all five programs	Pass
TC-10	Verify super.display() chaining in all files	Each level of display() calls the parent level, producing cumulative output lines	Output lines accumulated correctly through the prototype chain in all programs	Pass

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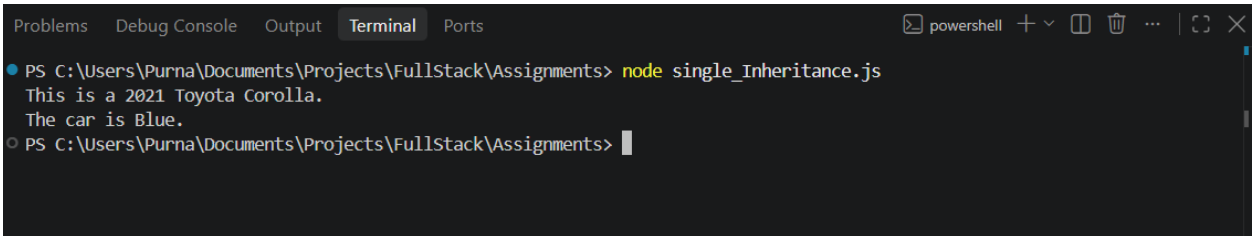
9. OUTPUT

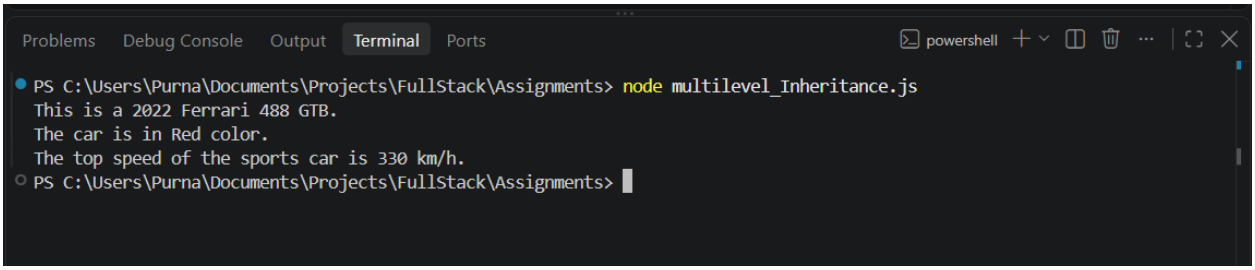
Output 1: Single Inheritance

Running single_Inheritance.js prints two lines: "This is a 2021 Toyota Corolla." from the Vehicle display() called via super, followed by "The car is in Blue color." from the Car display().



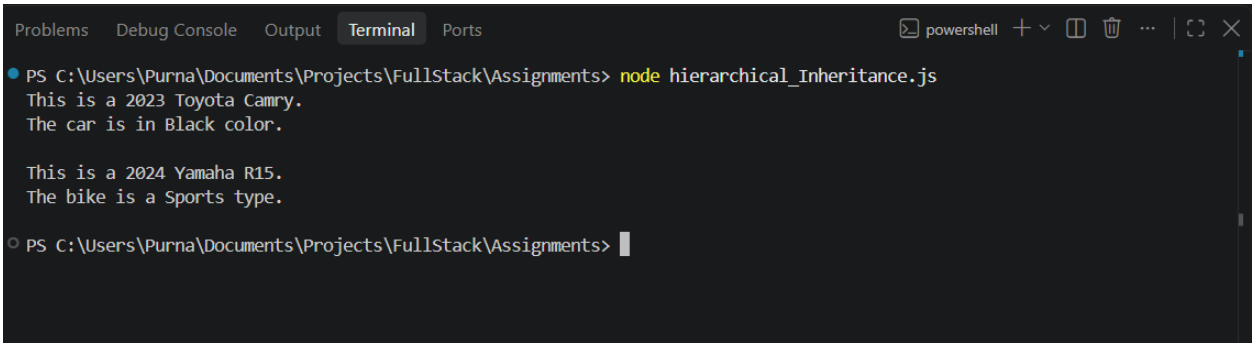
Output 2: Multilevel Inheritance

Running multilevel_Inheritance.js prints three chained lines for the SportsCar object: the Vehicle line, the Car colour line, and the SportsCar top speed line, demonstrating the full three-level display() chain.



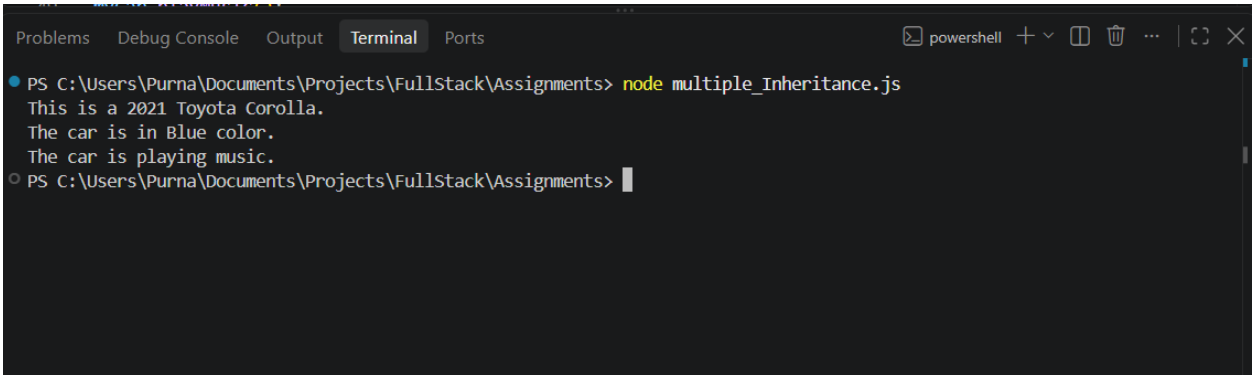
Output 3: Hierarchical Inheritance

Running hierarchical_Inheritance.js prints two separate blocks. The Car object prints the Vehicle line and its colour. The Bike object prints the Vehicle line and its type, showing two independent child classes sharing the same base.



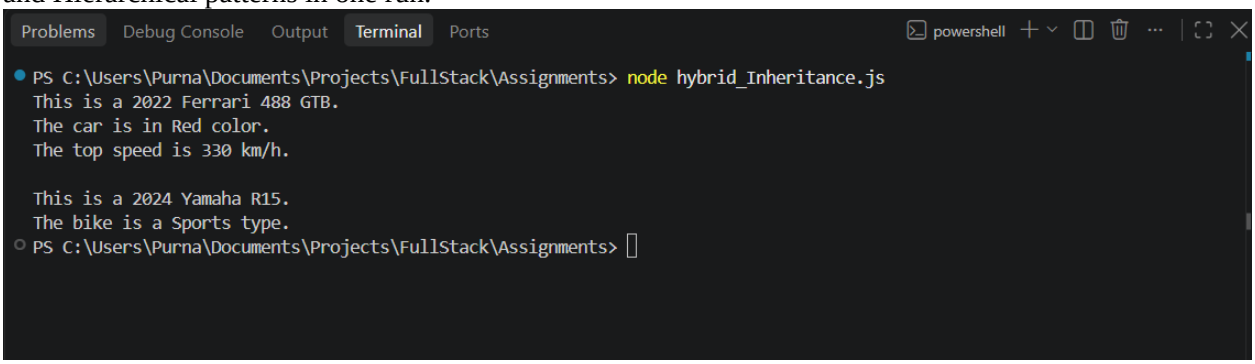
Output 4: Multiple Inheritance (Mixin)

Running multiple_Inheritance.js prints the Vehicle and Car display lines via display(), then "The car is playing music." via playMusic(), confirming the Car object has access to both the Vehicle prototype chain and the MusicMixin method.



Output 5: Hybrid Inheritance

Running hybrid_Inheritance.js prints the full three-level chain (Vehicle, Car, SportsCar) for the SportsCar object, followed by a separate two-level output (Vehicle, Bike) for the Bike object, combining Multilevel and Hierarchical patterns in one run.



10. RESULT

Thus, all five types of inheritance — Single, Multilevel, Hierarchical, Multiple (via Mixins), and Hybrid — were successfully implemented in JavaScript using ES6 class syntax. Each program demonstrates the use of the extends keyword, super() constructor chaining, method overriding, and super.display() to produce cumulative output. All ten test cases passed successfully, confirming correct inheritance behaviour, property initialisation, and method chaining across all five programs.